

Dynamics of Computation in the Brain  
76908  
Solution EX #4

Barak Haim

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# 1 Analytical Exercise: Ring model with a complex input

For  $i = 0, \dots, N-1$ :  $\theta_i = \frac{2\pi i}{N}$ .  $\tilde{J}_{ij} = \frac{1}{N} \tilde{J}(\theta_i - \theta_j)$ .

$$\begin{aligned} \frac{dr_i(t)}{dt} &= -r_i(t) + \sum J_{ij} r_j + h_i \\ h_i &= \begin{cases} 2 & |\pi - \theta_i| > \frac{\pi}{2} \\ 0 & \text{else} \end{cases} \\ \tilde{J}(\theta) &= \begin{cases} J & |\theta| < \frac{\pi}{2} \text{ or } \frac{3\pi}{2} < |\theta| \\ 0 & \text{else} \end{cases} \end{aligned}$$

## 1.1 $N \rightarrow \infty$

Just as we did in class, we write in matrix notation the continuous some over the ring (from  $-\pi$  to  $\pi$ ):

$$\begin{aligned} \dot{r}(t, \theta) &= -r(\theta, t) + \frac{1}{2\pi} \int_{-\pi}^{\pi} \tilde{J}(\theta - \varphi) r(\varphi, t) d\varphi + h(\theta, t) \\ &= -r(\theta, t) + \frac{1}{2\pi} \int_{\theta - \frac{\pi}{2}}^{\theta + \frac{\pi}{2}} J r(\varphi, t) d\varphi + h(\theta, t) \end{aligned}$$

## 1.2 Fourier decomposition of $r$ and $h$

Using chapter #3, eq- 3.12-18, First:

$$r(\theta, t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n(t) \cos(n\theta) + b_n(t) \sin(n\theta)]$$

Where:

$$a_n(t) = \frac{1}{\pi} \int_{-\pi}^{\pi} r(\theta, t) \cos(n\theta) d\theta \quad \& \quad b_n(t) = \frac{1}{\pi} \int_{-\pi}^{\pi} r(\theta, t) \sin(n\theta) d\theta$$

We go back to the other parts of the sum in 1.1, we'll look at their Fourier decomposition separately as we did for  $r$ . Once we'll put them all together, we can compare them with the Fourier decomposition of  $\dot{r}$  and get a set of decoupled dynamical equations. Moving on to the second member in the sum, the varying part of the

input:

$$\begin{aligned}
\frac{1}{2\pi} \int_{\theta-\frac{\pi}{2}}^{\theta+\frac{\pi}{2}} Jr(\varphi, t) d\varphi &= \frac{1}{2\pi} \int_{\theta-\frac{\pi}{2}}^{\theta+\frac{\pi}{2}} J \left[ \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n(t) \cos(n\varphi) + b_n(t) \sin(n\varphi)] \right] dv\theta \\
&= \frac{J}{2\pi} \left[ \int_{\theta-\frac{\pi}{2}}^{\theta+\frac{\pi}{2}} \frac{a_0}{2} d\varphi + \sum_{n=1}^{\infty} a_n \int_{\theta-\frac{\pi}{2}}^{\theta+\frac{\pi}{2}} \cos(n\varphi) d\varphi + \sum_{n=1}^{\infty} b_n \int_{\theta-\frac{\pi}{2}}^{\theta+\frac{\pi}{2}} \sin(n\varphi) d\varphi \right] \\
&= \frac{J}{2\pi} \left[ \int_{\theta-\frac{\pi}{2}}^{\theta+\frac{\pi}{2}} \frac{a_0}{2} d\varphi + \sum_{n=1}^{\infty} a_n \frac{\sin(n\varphi)}{n} \Big|_{\theta-\frac{\pi}{2}}^{\theta+\frac{\pi}{2}} - \sum_{n=1}^{\infty} b_n \frac{\cos(n\varphi)}{n} \Big|_{\theta-\frac{\pi}{2}}^{\theta+\frac{\pi}{2}} \right] \\
&= \frac{J}{2\pi} \int_{\theta-\frac{\pi}{2}}^{\theta+\frac{\pi}{2}} \frac{a_0}{2} d\varphi \\
&\quad + \frac{J}{2\pi} \sum_{n=1}^{\infty} \frac{a_n}{n} \left[ \sin \left( n\theta + \frac{n\pi}{2} \right) - \sin \left( n\theta - \frac{n\pi}{2} \right) \right] \\
&\quad - \frac{J}{2\pi} \sum_{n=1}^{\infty} \frac{b_n}{n} \left[ \cos \left( n\theta + \frac{n\pi}{2} \right) - \cos \left( n\theta - \frac{n\pi}{2} \right) \right] \\
&= \frac{J}{2\pi} \left[ \int_{\theta-\frac{\pi}{2}}^{\theta+\frac{\pi}{2}} \frac{a_0}{2} d\varphi + \sum_{n=1}^{\infty} \frac{a_n}{n} \left[ 2\cos(n\theta) \sin \left( \frac{n\pi}{2} \right) \right] - \sum_{n=1}^{\infty} \frac{b_n}{n} \left[ -2\sin(n\theta) \sin \left( \frac{n\pi}{2} \right) \right] \right] \\
&= \frac{J}{2\pi} \left[ \frac{a_0}{2} \left[ \theta + \frac{\pi}{2} - \theta + \frac{\pi}{2} \right] + \sum_{n=1}^{\infty} \frac{a_n}{n} \left[ 2\cos(n\theta) \sin \left( \frac{n\pi}{2} \right) \right] + \sum_{n=1}^{\infty} \frac{b_n}{n} \left[ 2\sin(n\theta) \sin \left( \frac{n\pi}{2} \right) \right] \right] \\
&= \frac{J}{2\pi} \left[ \frac{a_0\pi}{2} + \sum_{n=1}^{\infty} \frac{a_n}{n} \left[ 2\cos(n\theta) \sin \left( \frac{n\pi}{2} \right) \right] + \sum_{n=1}^{\infty} \frac{b_n}{n} \left[ 2\sin(n\theta) \sin \left( \frac{n\pi}{2} \right) \right] \right] \\
&= \frac{Ja_0}{4} + \frac{J}{2\pi} \sum_{n=1}^{\infty} \left[ \frac{a_n}{n} \left[ 2\cos(n\theta) \sin \left( \frac{n\pi}{2} \right) \right] + \frac{b_n}{n} \left[ 2\sin(n\theta) \sin \left( \frac{n\pi}{2} \right) \right] \right] \\
&= \frac{Ja_0}{4} + \sum_{n=1}^{\infty} \frac{J}{\pi n} \sin \left( \frac{n\pi}{2} \right) [a_n \cos(n\theta) + b_n \sin(n\theta)] \\
&= \frac{Ja_0}{4} + \sum_{n=1}^{\infty} \left[ \frac{J}{\pi n} \sin \left( \frac{n\pi}{2} \right) a_n \right] \cos(n\theta) + \sum_{n=1}^{\infty} \left[ \frac{J}{\pi n} \sin \left( \frac{n\pi}{2} \right) b_n \right] \sin(n\theta)
\end{aligned}$$

Where equality #1 (explained in the TA, page #5) uses the identities:

$$\begin{aligned}
\sin \left( n\theta \pm \frac{n\pi}{2} \right) &= \sin(n\theta) \cos \left( \frac{n\pi}{2} \right) \pm \sin(n\theta) \cos \left( \frac{n\pi}{2} \right) \\
\cos \left( n\theta \pm \frac{n\pi}{2} \right) &= \cos(n\theta) \cos \left( \frac{n\pi}{2} \right) \mp \sin(n\theta) \sin \left( \frac{n\pi}{2} \right)
\end{aligned}$$

And thus:

$$\begin{aligned}
\sin \left( n\theta + \frac{n\pi}{2} \right) - \sin \left( n\theta - \frac{n\pi}{2} \right) &= \sin(n\theta) \cos \left( \frac{n\pi}{2} \right) + \sin(n\theta) \cos \left( \frac{n\pi}{2} \right) - \sin(n\theta) \cos \left( \frac{n\pi}{2} \right) + \sin(n\theta) \cos \left( \frac{n\pi}{2} \right) \\
&= 2\cos(n\theta) \sin \left( \frac{n\pi}{2} \right)
\end{aligned}$$

and:

$$\begin{aligned}\cos\left(n\theta + \frac{n\pi}{2}\right) - \cos\left(n\theta - \frac{n\pi}{2}\right) &= \cos(n\theta)\cos\left(\frac{n\pi}{2}\right) - \sin(n\theta)\sin\left(\frac{n\pi}{2}\right) - \cos(n\theta)\cos\left(\frac{n\pi}{2}\right) - \sin(n\theta)\sin\left(\frac{n\pi}{2}\right) \\ &= -2\sin(n\theta)\sin\left(\frac{n\pi}{2}\right)\end{aligned}$$

We got the Fourier components of the varying part of the input. Now moving on to constant part of the input (note there may arise a slight confusion here from the notation, usually we refer to all of the neural input as  $h$  but here only to the fixed external component):

$$h(\theta) = 1 + f(\theta) = 1 + \begin{cases} -1 & -\pi < \theta < -\frac{\pi}{2} \\ 1 & -\frac{\pi}{2} < \theta < \frac{\pi}{2} \\ -1 & \frac{\pi}{2} < \theta < \pi \end{cases}$$

for  $f(\theta)$  from exercise #2, question 1.1. We've already calculated its Fourier decomposition:

$$\begin{aligned}h(\theta) &= 1 + \frac{4}{\pi n} \left( \sum_{n=1,5,9,\dots} \cos(n\theta) - \sum_{n=3,7,11,\dots} \cos(n\theta) \right) = 1 + \frac{4}{\pi n} \sum_{n=1,3,5,\dots} (-1)^{\frac{n-1}{2}} \cos(n\theta) \\ &= 1 + \sum_{n=1}^{\infty} \left[ \frac{4}{\pi n} \sin\left(\frac{n\pi}{2}\right) \right] \cos(n\theta) = \frac{2}{2} + \sum_{n=1}^{\infty} \left[ \frac{4}{\pi n} \sin\left(\frac{n\pi}{2}\right) \right] \cos(n\theta) + \sum_{n=1}^{\infty} 0 \cdot \sin(n\theta)\end{aligned}$$

Putting it all together, we join the coefficients of the sine, cosine and free members together and get:

$$\begin{aligned}\dot{r} &= -r + \frac{1}{2\pi} \int_{\theta-\frac{\pi}{2}}^{\theta+\frac{\pi}{2}} Jr(\varphi)d\varphi + h(\theta) \\ &= -\frac{a_0}{2} + \sum_{n=1}^{\infty} [-a_n](t)\cos(n\theta) + \sum_{n=1}^{\infty} [-b_n](t)\sin(n\theta) \\ &\quad + \frac{Ja_0}{4} + \sum_{n=1}^{\infty} \left[ \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) a_n \right] \cos(n\theta) + \sum_{n=1}^{\infty} \left[ \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) b_n \right] \sin(n\theta) \\ &\quad + 1 + \sum_{n=1}^{\infty} \left[ \frac{4}{\pi n} \sin\left(\frac{n\pi}{2}\right) \right] \cos(n\theta) \\ &= \left[ -\frac{a_0}{2} + \frac{Ja_0}{4} + 1 \right] + \\ &\quad + \sum_{n=1}^{\infty} \left[ \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) a_n - a_n + \frac{4}{\pi n} \sin\left(\frac{n\pi}{2}\right) \right] \cos(n\theta) \\ &\quad + \sum_{n=1}^{\infty} \left[ \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) b_n - b_n \right] \sin(n\theta) \\ &= \frac{(J-2)a_0 + 4}{4} + \sum_{n=1}^{\infty} \left[ \left( \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) - 1 \right) a_n + \frac{4}{\pi n} \sin\left(\frac{n\pi}{2}\right) \right] \cos(n\theta) + \sum_{n=1}^{\infty} \left[ \left( \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) - 1 \right) b_n \right] \sin(n\theta)\end{aligned}$$

What we got defines the Fourier decomposition of  $\dot{r}$  in the terms of  $a_0, a_n$  &  $b_n$ . Now, We also know that:

$$\begin{aligned}
\dot{r} &= \frac{dr}{dt}(\theta, t) = \frac{d}{dt} \frac{a_0(t)}{2} + \frac{d}{dt} \sum_{n=1}^{\infty} a_n(t) \cos(n\theta) + \frac{d}{dt} \sum_{n=1}^{\infty} b_n(t) \sin(n\theta) \\
&= \frac{\dot{a}_0}{2} + \sum_{n=1}^{\infty} \frac{d}{dt} (a_n(t) \cos(n\theta)) + \sum_{n=1}^{\infty} \frac{d}{dt} (b_n(t) \sin(n\theta)) \\
&= \frac{\dot{a}_0}{2} + \sum_{n=1}^{\infty} \left( \cos(n\theta) \frac{d}{dt} a_n(t) + a_n(t) \frac{d}{dt} \cos(n\theta) \right) + \sum_{n=1}^{\infty} \left( \sin(n\theta) \frac{d}{dt} b_n(t) + b_n(t) \frac{d}{dt} \sin(n\theta) \right) \\
&= \frac{\dot{a}_0}{2} + \sum_{n=1}^{\infty} \dot{a}_n(t) \cos(n\theta) + \sum_{n=1}^{\infty} \dot{b}_n(t) \sin(n\theta)
\end{aligned}$$

Where the last equality is holds because  $\sin(n\theta)$  and  $\cos(n\theta)$  do not change with respect to time and so  $0 = \frac{d}{dt} \sin(n\theta) = \frac{d}{dt} \cos(n\theta)$ . What we got is two different ways to write the Fourier decomposition of  $\dot{r}$ . We can compare the coefficients element-wise (which is possible as we were dealing with a linear combination of eigenfunctions) and get a set of decoupled ODE's:

$$\begin{aligned}
\frac{\dot{a}_0}{2} &= \frac{(J-2)a_0 + 4}{4} \implies \dot{a}_0 = \frac{(J-2)a_0 + 4}{2} = a_0 \left( \frac{J}{2} - 1 \right) + 2 \\
\dot{a}_n &= \left( \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) - 1 \right) a_n + \frac{4}{\pi n} \sin\left(\frac{n\pi}{2}\right) \\
\dot{b}_n &= \left( \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) - 1 \right) b_n
\end{aligned}$$

### 1.3 Fixed point analysis

We want to find when  $\dot{a}_0 = \dot{a}_n = \dot{b}_n = 0$ :

$$\begin{cases} 0 \stackrel{!}{=} \frac{\dot{a}_0}{2} = a_0 \left( \frac{J}{2} - 1 \right) + 2 \implies & a_0^* = -\frac{2}{\left(\frac{J}{2}-1\right)} = \frac{-4}{J-2} = \frac{4}{2-J} \\ 0 \stackrel{!}{=} \dot{a}_n = \left( \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) - 1 \right) a_n + \frac{4}{\pi n} \sin\left(\frac{n\pi}{2}\right) \implies & a_n^* = \frac{\frac{4}{\pi n} \sin\left(\frac{n\pi}{2}\right)}{\left(1 - \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right)\right)} = \frac{4 \sin\left(\frac{n\pi}{2}\right)}{\pi n - J \sin\left(\frac{n\pi}{2}\right)} \\ 0 \stackrel{!}{=} \dot{b}_n = \left( \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) - 1 \right) b_n \implies & b_n^* = 0 \end{cases}$$

Note that if  $\frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) = 1$  then we can have  $b_n^* \neq 0$ . This is iff  $\sin\left(\frac{n\pi}{2}\right) = \frac{\pi n}{J}$  and may only hold for a single  $n$  ( $\text{mod}_4$  1) and a suitable  $J$ .

### 1.4 Fixed point stability

As  $b_n = 0$  at the FP's we can ignore it and only look at the eigenvalues of the linearized systems of  $a_n$ . As for  $a_0$ :

$$\frac{d\dot{a}_0}{da_0} = \frac{J}{2} - 1 < 0 \iff J < 2$$

As for  $a_n$  for even  $n$ 's:  $\sin\left(\frac{n\pi}{2}\right) = 0$  so:

$$\frac{d\dot{a}_n}{da_n} = \frac{J}{\pi n} \sin\left(\frac{n\pi}{2}\right) - 1 = -1 < 0$$

And thus, those coefficients would converge. For  $n = 3, 7, 11, \dots$  we get:  $\sin(\frac{n\pi}{2}) = -1$ , so:

$$\frac{d\dot{a}_n}{da_n} = \frac{J}{\pi n} \sin(\frac{n\pi}{2}) - 1 = -\frac{J}{\pi n} - 1 < 0$$

as  $\frac{J}{\pi n}$  is a positive number. Lastly we're left with the case when  $n = 1, 5, 9, \dots$ . In this case  $\sin(\frac{n\pi}{2}) = 1$  and:

$$\frac{d\dot{a}_n}{da_n} = \frac{J}{\pi n} \sin(\frac{n\pi}{2}) - 1 = \frac{J}{\pi n} - 1 < 0 \iff \frac{J}{\pi n} < 1 \iff J < \pi n \iff J < \pi < 5\pi < 9\pi \dots$$

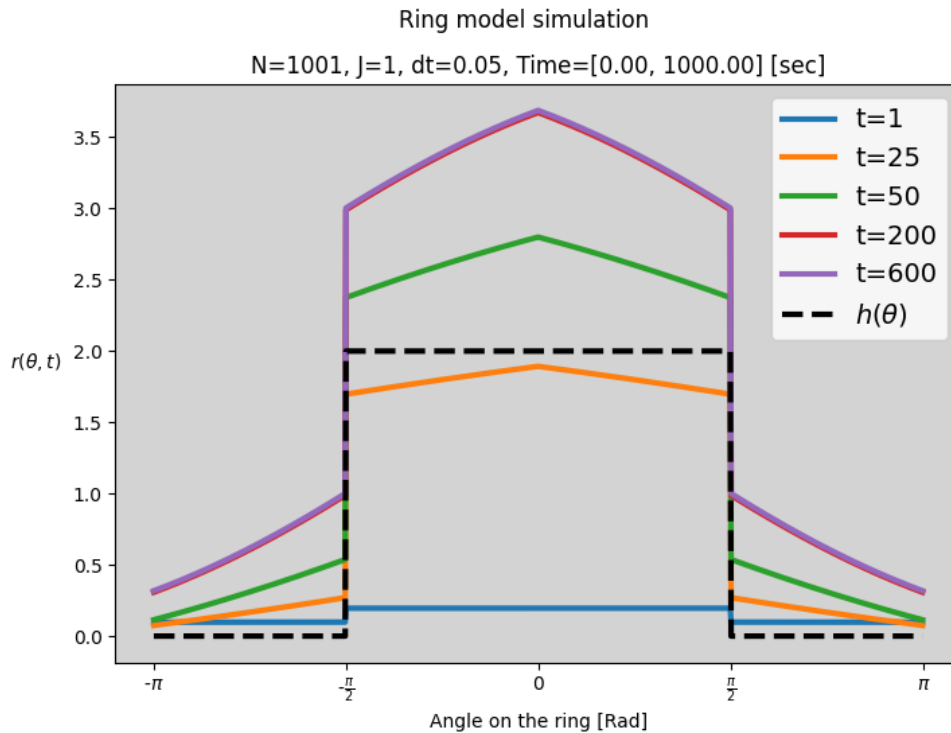
So as  $n \rightarrow \infty$ , we only need to check for the case  $n = 1$  and we know the condition for bigger  $n$ 's will be met. As we've already demanded for the  $a_0$  case that  $J < 2 < \pi$  and the condition that  $\frac{d\dot{a}_n}{da_n} < 0$  will be satisfied. In conclusion, we get stable fixed points iff:

$$J < 2$$

## 2 Numerical Exercise: The ring model

### 2.1 Simulating the ring model

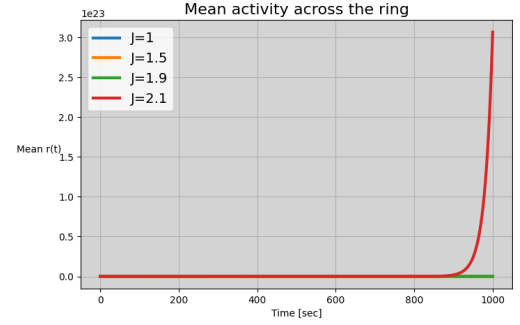
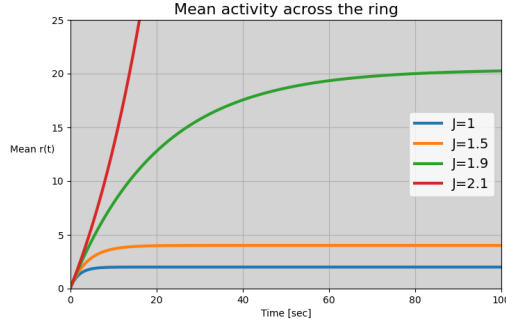
Here's a plot of the activation on the ring at different times by position on the ring (centered around 0 and ranging from  $-\pi$  to  $\pi$ ). The simulation took about 2.65 to complete. We see the bump in the center of the ring rises with time and converges to its fixed point.



## 2.2 Studying the effect of J

### 2.2.1 Mean firing rate for $J = 1, 1.5, 1.9, 2.1$

We can see that for  $J = 2.1$  the dynamics diverges. Zooming in, we can how quickly the dynamics converges for  $J < 2$ :



Mean activity VS time zoom in      Mean activity VS time, for  $J = 2.1$  we get divergences  
From these graphs we see that for  $J < 2$  the dynamics converges to a fixed point relatively quickly compared to the overall run time.

### 2.2.2 For a general function $f$ , $a_0$ is twice its average

By definition:

$$f_{avg} := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2} \left( \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \right) = \frac{1}{2} a_0$$

### 2.2.3 The ODE derived for $a_0$ determines the FP and rat of convergence

We've computed the equation for  $a_0$  and the fixed point derived by it:

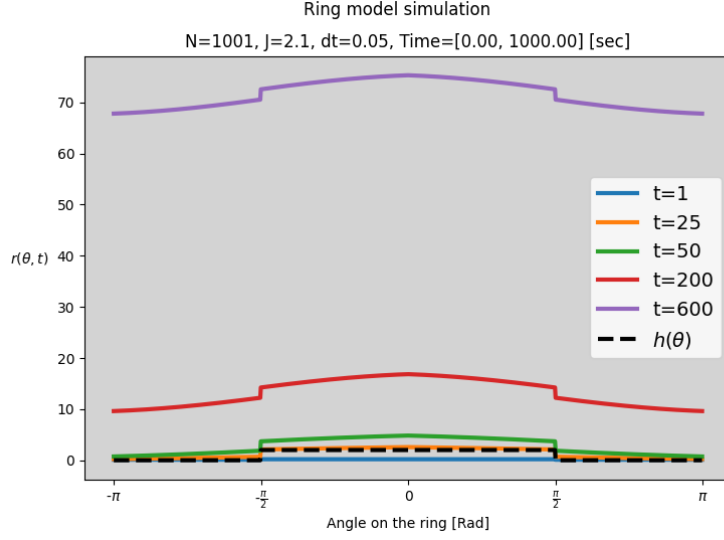
$$\dot{a}_0 = \left( \frac{J}{2} - 1 \right) a_0 + 2$$

$$a_0^* = \frac{4}{2 - J}$$

From this we can tell that when  $J \nearrow 2$ , the value of  $(2 - J) \searrow 0$  and thus  $a_0^* \nearrow \infty$ . I.e. the average firing rate at the FP would be greater. Also, when  $J \nearrow 2$ , the value of  $\left(\frac{J}{2} - 1\right) \nearrow 0$ . I.e. the rate of change for the average firing rate is greater. Eventually, when  $2 < J$ ,  $0 < \left(\frac{J}{2} - 1\right)$  and the average firing rate goes to infinity

### 2.2.4 $J = 2.1$ Only $a_0$ diverges

As showed in 1.4, the  $b_n$ 's are always 0. The  $a_n$ 's converge when  $J < \pi$  but if  $2 < J$  then  $a_0$  diverges.



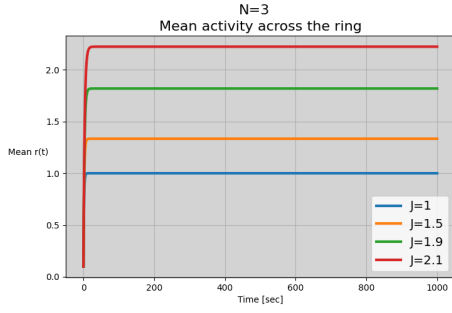
Firing rate as a function of position on the ring [rad] at different time-steps

Here we see plots of the firing rate on the ring for different time points when  $2 < J$ . We can see how the the structure is kept but everything moves upwards together.

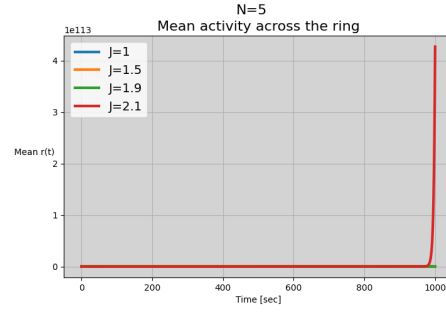
### 3 Qualitative Observations

#### 3.1 When is N “large enough”

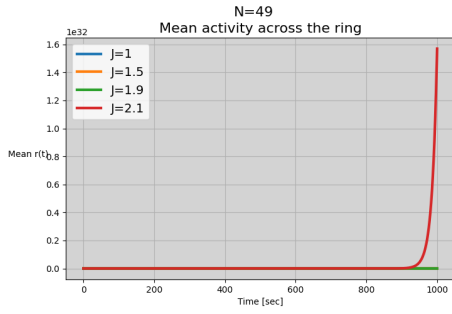
For me, generally speaking, if a number is larger than the number of my fingers ( $\approx 10$ ), I consider it to be infinity. For some people, nothing is ever large enough... In our context, the question boils down to how well the discrete sum  $\sum J_{ij}r_j$  approximates the integral  $\frac{1}{2\pi} \int_{-\pi}^{\pi} \tilde{J}(\theta - \varphi)r(\varphi, t)d\varphi$ . This is a function of  $N$ ,  $dt$  and maybe more (like the method of numerical solver). Basically, as  $N$  is larger,  $\Delta\theta = \frac{2\pi}{N}$  is smaller which makes the discrete sum closer to the integral. From other simulations I ran, I found that in my case when  $N = 3$  the simulation does not diverge for  $J = 2.1$ , however, for  $5 \leq N$ , it does (as could be seen in the graphs below). Generally, I would choose  $N$  such that  $\Delta\theta$  is sensitive enough to changes in the different regimes of the ring. Which in our case is  $\Delta\theta < \frac{\pi}{2}$ . Even better is  $\Delta\theta \ll 1$ . All in all, we can see the choice of  $N = 1001$ , in this particular case, is “good enough”.



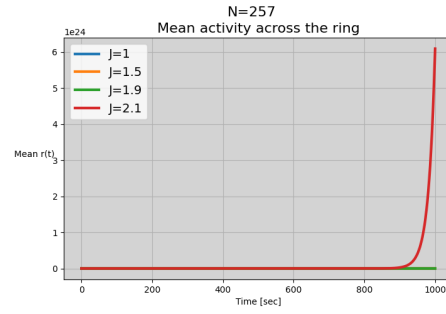
N=3 - Wrongful convergence



N=5



N=49



N=257

### 3.2 External input as well as connectivity profile affect the dynamics

The external input determines the excited part of the ring where's the connectivity matrix determines the peak and trough of the firing profile. As we can see in section 2.1, the bump peaks at the center of the ring as the neuron at 0 gets the most input - it gets the external input and the maximal amount possible of input from other neurons around the half ring. On the other hand, the neuron antipodal to it, gets no external input and the minimal amount possible of input from other neurons around its half ring. We can see the most "light up" part is the part which gets strictly positive external input. Even more so, this is the most connected part on the ring.